ADI-FDTD for Inhomogeneous Media

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1 Preliminaries

1.1 Maxwell's Equations

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} \tag{1.1a}$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \tag{1.1b}$$

Expanding (1.1a) into components:

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \varepsilon \frac{\partial E_x}{\partial t} + \sigma E_x \tag{1.2a}$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = \varepsilon \frac{\partial E_y}{\partial t} + \sigma E_y \tag{1.2b}$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \varepsilon \frac{\partial E_z}{\partial t} + \sigma E_z \tag{1.2c}$$

Expanding (1.1b) into components:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \right) \tag{1.3a}$$

$$\frac{\partial H_y}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \right) \tag{1.3b}$$

$$\frac{\partial H_z}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \right) \tag{1.3c}$$

1.2 Yee Grid in AMReX

The Yee grid stores different field components at different physical locations within each computational cell. In AMReX this staggering is represented by the `IndexType` used to define each `MultiFab`: a cell-centered direction has index type 0 and is located at $i + \frac{1}{2}$, while a nodal direction has index type 1 and is located at i. The implementation uses

$$\mathbf{E}_\alpha : \text{cell-centered in the } \alpha \text{ direction and nodal in the other two directions,}$$
$$\mathbf{H}_\alpha : \text{nodal in the } \alpha \text{ direction and cell-centered in the other two directions.}$$

Thus the component locations are

Component	x location	y location	z location
$E_{x,i,j,k}$	$i + \frac{1}{2}$	j	k
$E_{y,i,j,k}$	i	$j + \frac{1}{2}$	k
$E_{z,i,j,k}$	i	j	$k + \frac{1}{2}$
$H_{x,i,j,k}$	i	$j + \frac{1}{2}$	$k + \frac{1}{2}$
$H_{y,i,j,k}$	$i + \frac{1}{2}$	j	$k + \frac{1}{2}$
$H_{z,i,j,k}$	$i + \frac{1}{2}$	$j + \frac{1}{2}$	k

The half-cell offsets therefore do not appear explicitly as array indices such as $i+\frac{1}{2}$. They are encoded in the `MultiFab` staggering. The integer offsets in the finite-difference stencil choose the two source values that bracket the field being updated. For example, $H_{x,i,j,k}$ is located at $j+\frac{1}{2}$ in the y direction, while $E_{z,i,j,k}$ and $E_{z,i,j+1,k}$ are located at j and $j+1$. Therefore

$$\left.\frac{\partial E_z}{\partial y}\right|_{H_{x,i,j,k}} \approx \frac{E_{z,i,j+1,k} - E_{z,i,j,k}}{\Delta y}. \tag{1.4}$$

Similarly, $E_{x,i,j,k}$ is located at j, while $H_{z,i,j-1,k}$ and $H_{z,i,j,k}$ are located at $j-\frac{1}{2}$ and $j+\frac{1}{2}$, so

$$\left.\frac{\partial H_z}{\partial y}\right|_{E_{x,i,j,k}} \approx \frac{H_{z,i,j,k} - H_{z,i,j-1,k}}{\Delta y}. \tag{1.5}$$

This is why the magnetic update below uses forward differences such as $j+1$ and $k+1$, while the electric update uses backward differences such as $j-1$ and $k-1$: each derivative is centered at the staggered location of the field component being updated.

2 FDTD Method

The explicit Yee FDTD update used here is written in three stages, matching the implementation: first update $\mathbf{H}$ by a half step, then update $\mathbf{E}$ by a full step, and finally update $\mathbf{H}$ by another half step. Define the update coefficients

$$C_a = \frac{2\varepsilon - \sigma\Delta t}{2\varepsilon + \sigma\Delta t}, \qquad C_b = \frac{2\Delta t}{2\varepsilon + \sigma\Delta t}, \qquad D_b = \frac{\Delta t}{\mu}, \tag{2.1}$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity.

2.1 First Magnetic Half-Step

$$H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{2}\left[\frac{E^n_{z,i,j+1,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}}{\Delta y} - \frac{E^n_{y,i,j+\frac{1}{2},k+1} - E^n_{y,i,j+\frac{1}{2},k}}{\Delta z}\right] \tag{2.2a}$$

$$H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{2}\left[\frac{E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta z} - \frac{E^n_{z,i+1,j,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}}{\Delta x}\right] \tag{2.2b}$$

$$H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{2}\left[\frac{E^n_{y,i+1,j+\frac{1}{2},k} - E^n_{y,i,j+\frac{1}{2},k}}{\Delta x} - \frac{E^n_{x,i+\frac{1}{2},j+1,k} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta y}\right]. \tag{2.2c}$$

In the AMReX Yee grid, with vacuum media, the magnetic field updates are implemented as

$$B_x(i,j,k) = B_x(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_z(i,j+1,k) - E_z(i,j,k)}{\Delta y} - \frac{E_y(i,j,k+1) - E_y(i,j,k)}{\Delta z}\right] \tag{2.3a}$$

$$B_y(i,j,k) = B_y(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_x(i,j,k+1) - E_x(i,j,k)}{\Delta z} - \frac{E_z(i+1,j,k) - E_z(i,j,k)}{\Delta x}\right] \tag{2.3b}$$

$$B_z(i,j,k) = B_z(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_y(i+1,j,k) - E_y(i,j,k)}{\Delta x} - \frac{E_x(i,j+1,k) - E_x(i,j,k)}{\Delta y}\right]. \tag{2.3c}$$

2.2 Electric Full-Step

$$E^{n+1}_{x,i+\frac{1}{2},j,k} = C^{n+1}_{a,i+\frac{1}{2},j,k}E^{n}_{x,i+\frac{1}{2},j,k} + C^{n+1}_{b,i+\frac{1}{2},j,k}\left[\frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}\right] \tag{2.4a}$$

$$E^{n+1}_{y,i,j+\frac{1}{2},k} = C^{n+1}_{a,i,j+\frac{1}{2},k}E^{n}_{y,i,j+\frac{1}{2},k} + C^{n+1}_{b,i,j+\frac{1}{2},k}\left[\frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}\right] \tag{2.4b}$$

$$E^{n+1}_{z,i,j,k+\frac{1}{2}} = C^{n+1}_{a,i,j,k+\frac{1}{2}}E^{n}_{z,i,j,k+\frac{1}{2}} + C^{n+1}_{b,i,j,k+\frac{1}{2}}\left[\frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}\right]. \tag{2.4c}$$

In the AMReX Yee grid, with vacuum media, the electric field updates are implemented as

$$E_x(i,j,k) = E_x(i,j,k) + c^2\Delta t\left[\frac{B_z(i,j,k) - B_z(i,j-1,k)}{\Delta y} - \frac{B_y(i,j,k) - B_y(i,j,k-1)}{\Delta z}\right] \tag{2.5a}$$

$$E_y(i,j,k) = E_y(i,j,k) + c^2\Delta t\left[\frac{B_x(i,j,k) - B_x(i,j,k-1)}{\Delta z} - \frac{B_z(i,j,k) - B_z(i-1,j,k)}{\Delta x}\right] \tag{2.5b}$$

$$E_z(i,j,k) = E_z(i,j,k) + c^2\Delta t\left[\frac{B_y(i,j,k) - B_y(i-1,j,k)}{\Delta x} - \frac{B_x(i,j,k) - B_x(i,j-1,k)}{\Delta y}\right]. \tag{2.5c}$$

2.3 Second Magnetic Half-Step

$$H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{2}\left[\frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y} - \frac{E^{n+1}_{y,i,j+\frac{1}{2},k+1} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta z}\right] \tag{2.6a}$$

$$H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{2}\left[\frac{E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta z} - \frac{E^{n+1}_{z,i+1,j,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta x}\right] \tag{2.6b}$$

$$H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{2}\left[\frac{E^{n+1}_{y,i+1,j+\frac{1}{2},k} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j+1,k} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta y}\right]. \tag{2.6c}$$

In the AMReX Yee grid, with vacuum media, the second magnetic field updates are implemented as

$$B_x(i,j,k) = B_x(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_z(i,j+1,k) - E_z(i,j,k)}{\Delta y} - \frac{E_y(i,j,k+1) - E_y(i,j,k)}{\Delta z}\right] \tag{2.7a}$$

$$B_y(i,j,k) = B_y(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_x(i,j,k+1) - E_x(i,j,k)}{\Delta z} - \frac{E_z(i+1,j,k) - E_z(i,j,k)}{\Delta x}\right] \tag{2.7b}$$

$$B_z(i,j,k) = B_z(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_y(i+1,j,k) - E_y(i,j,k)}{\Delta x} - \frac{E_x(i,j+1,k) - E_x(i,j,k)}{\Delta y}\right]. \tag{2.7c}$$

This explicit method is simple and local, but the time step is restricted by the CFL stability condition. For a uniform lossless grid, a typical stability limit is

$$\Delta t \le \frac{1}{c\sqrt{\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2}}}, \qquad c = \frac{1}{\sqrt{\mu\varepsilon}}. \tag{2.8}$$

3 ADI Method

Here we derive the ADI-FDTD formulation in [1].
The ADI-FDTD update advances the fields by a full time step Δt in two half steps of size $\Delta t/2$. Unlike the explicit three-stage leapfrog update above, each ADI half step first updates $\mathbf{E}$ implicitly over $\Delta t/2$, then updates $\mathbf{H}$ explicitly over $\Delta t/2$ from the updated electric field. The first half step treats one set of implicit directions; the second half step treats the complementary directions to complete the cycle. Define the half-step update coefficients

$$C_a = \frac{4\varepsilon - \sigma\Delta t}{4\varepsilon + \sigma\Delta t}, \qquad C_b = \frac{2\Delta t}{4\varepsilon + \sigma\Delta t}, \qquad D_b = \frac{\Delta t}{2\mu}, \tag{3.1}$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity. These differ from the full-step coefficients in (2.1) because each electric update uses $\Delta t/2$.

3.1 First ADI Half-Step

E_x and H_z By equation (1.2a),

$$\varepsilon_{i+\frac{1}{2},j,k}\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta t/2} + \sigma_{i+\frac{1}{2},j,k}\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + E^n_{x,i+\frac{1}{2},j,k}}{2} = [H_1], \tag{3.2}$$

where

$$[H_1] = \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}. \tag{3.3}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_x^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_x^{n} + [H_1]. \tag{3.4}$$

Hence

$$E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k}\, [H_1], \tag{3.5}$$

By equation (1.3c),

$$\begin{aligned} H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \Bigg(& \frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} \\ & - \frac{E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta x} \Bigg), \end{aligned} \tag{3.6}$$

Substituting the magnetic update into (3.5) and eliminating $H_z^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} &= C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} \\ &\quad + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \Bigg[H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta x} \right) \\ &\qquad - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k} - D_{b,i+\frac{1}{2},j-\frac{1}{2},k} \left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k}}{\Delta y} - \frac{E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}}{\Delta x} \right) \Bigg] \\ &\quad - \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}} \right). \end{aligned} \tag{3.7}$$

Collecting the unknown $E_x^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} & - \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}\, E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} \\ & + \left[1 + \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} + \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2} \right] E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ & - \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}\, E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} \\ = {} & C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k} \right) \\ & - \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y\, \Delta x} \left(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k} \right) \\ & + \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y\, \Delta x} \left(E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k} \right) \\ & - \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}} \right). \end{aligned} \tag{3.8}$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} + \beta_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - \gamma_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} = p_{x1} & E^{n}_{x,i+\frac{1}{2},j,k} \\
& + \frac{1}{\Delta y}\Big(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\Big) \\
& - \frac{1}{\Delta z}\Big(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\Big) \\
& + \frac{q_{x1}}{\Delta x \Delta y}\Big(E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}\Big) \\
& - \frac{r_{x1}}{\Delta x \Delta y}\Big(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\Big),
\end{aligned} \tag{3.9}
$$

and the magnetic update can be written as

$$
H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + \frac{r_{x1}}{\Delta y}\Big(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\Big) - \frac{r_{x1}}{\Delta x}\Big(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\Big). \tag{3.10}
$$

where

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \tag{3.11}$$

$$\beta_{x1} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{3.12}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \tag{3.13}$$

$$p_{x1} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.14}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{3.15}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.16}$$

E_y **and** H_x By equation (1.2b),

$$
\varepsilon_{i,j+\frac{1}{2},k} \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k} \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} + E^{n}_{y,i,j+\frac{1}{2},k}}{2} = [H_2], \tag{3.17}
$$

where

$$
[H_2] = \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}. \tag{3.18}
$$

Rearranging gives

$$
\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_y^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_y^{n} + [H_2]. \tag{3.19}
$$

Hence

$$
E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} = C_{a,i,j+\frac{1}{2},k} E^{n}_{y,i,j+\frac{1}{2},k} + C_{b,i,j+\frac{1}{2},k} [H_2], \tag{3.20}
$$

By equation (1.3a),

$$
\begin{aligned}
H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \Bigg(& \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} \\
& - \frac{E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta y} \Bigg),
\end{aligned} \tag{3.21}
$$

Substituting the magnetic update into (3.20) and eliminating $H_x^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} &= C_{a,i,j+\frac{1}{2},k} E^{n}_{y,i,j+\frac{1}{2},k} \\
&\quad + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\Bigg[H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} - \frac{E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta y} \right) \\
&\qquad - H^{n}_{x,i,j+\frac{1}{2},k-\frac{1}{2}} - D_{b,i,j+\frac{1}{2},k-\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1}}{\Delta z} - \frac{E^{n}_{z,i,j+1,k-\frac{1}{2}} - E^{n}_{z,i,j,k-\frac{1}{2}}}{\Delta y} \right) \Bigg] \\
&\quad - \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k} \right).
\end{aligned} \tag{3.22}
$$

Collecting the unknown $E_y^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} \\
&+ \left[1 + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2} \right] E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \\
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1} \\
= {} & C_{a,i,j+\frac{1}{2},k} E^{n}_{y,i,j+\frac{1}{2},k} + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z} \left(H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n}_{x,i,j+\frac{1}{2},k-\frac{1}{2}} \right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z\,\Delta y} \left(E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}} \right) \\
&+ \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z\,\Delta y} \left(E^{n}_{z,i,j+1,k-\frac{1}{2}} - E^{n}_{z,i,j,k-\frac{1}{2}} \right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k} \right).
\end{aligned} \tag{3.23}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1} + \beta_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - \gamma_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} &= p_{y1} E^{n}_{y,i,j+\frac{1}{2},k} \\
&\quad + \frac{1}{\Delta z} \left(H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n}_{x,i,j+\frac{1}{2},k-\frac{1}{2}} \right) \\
&\quad - \frac{1}{\Delta x} \left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k} \right) \\
&\quad + \frac{q_{y1}}{\Delta y \Delta z} \left(E^{n}_{z,i,j+1,k-\frac{1}{2}} - E^{n}_{z,i,j,k-\frac{1}{2}} \right) \\
&\quad - \frac{r_{y1}}{\Delta y \Delta z} \left(E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}} \right),
\end{aligned} \tag{3.24}
$$

and the magnetic update can be written as

$$
H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + \frac{r_{y1}}{\Delta z} \left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \right) - \frac{r_{y1}}{\Delta y} \left(E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}} \right). \tag{3.25}
$$

where

$$\alpha_{y1} = \frac{D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2}, \tag{3.26}$$

$$\beta_{y1} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y1} + \gamma_{y1}, \tag{3.27}$$

$$\gamma_{y1} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2}, \tag{3.28}$$

$$p_{y1} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{3.29}$$

$$q_{y1} = D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}, \tag{3.30}$$

$$r_{y1} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{3.31}$$

E_z and H_y By equation (1.2c),

$$\varepsilon_{i,j,k+\frac{1}{2}} \frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}}{\Delta t/2} + \sigma_{i,j,k+\frac{1}{2}} \frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} + E^n_{z,i,j,k+\frac{1}{2}}}{2} = [H_3], \tag{3.32}$$

where

$$[H_3] = \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}. \tag{3.33}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_z^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_z^n + [H_3]. \tag{3.34}$$

Hence

$$E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} = C_{a,i,j,k+\frac{1}{2}}\, E^n_{z,i,j,k+\frac{1}{2}} + C_{b,i,j,k+\frac{1}{2}}\, [H_3], \tag{3.35}$$

By equation (1.3b),

$$\begin{aligned} H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \Bigg(& \frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} \\ & - \frac{E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta z}\Bigg), \end{aligned} \tag{3.36}$$

Substituting the magnetic update into (3.35) and eliminating $H_y^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} &= C_{a,i,j,k+\frac{1}{2}}\, E^n_{z,i,j,k+\frac{1}{2}} \\ &\quad + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \Bigg[H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta z} \right) \\ &\qquad - H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}} - D_{b,i-\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^n_{x,i-\frac{1}{2},j,k+1} - E^n_{x,i-\frac{1}{2},j,k}}{\Delta z} \right) \Bigg] \\ &\quad - \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}} \right). \end{aligned} \tag{3.37}$$

Collecting the unknown $E_z^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
&- \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} \\
&+ \left[1 + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} \right] E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \\
&- \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}} \\
= {} & C_{a,i,j,k+\frac{1}{2}} E^n_{z,i,j,k+\frac{1}{2}} + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}} \right) \\
&- \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x\, \Delta z} \left(E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k} \right) \\
&+ \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x\, \Delta z} \left(E^n_{x,i-\frac{1}{2},j,k+1} - E^n_{x,i-\frac{1}{2},j,k} \right) \\
&- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}} \right).
\end{aligned}
\tag{3.38}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{z1} E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}} + \beta_{z1} E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - \gamma_{z1} E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} = {} & p_{z1} E^n_{z,i,j,k+\frac{1}{2}} \\
&+ \frac{1}{\Delta x} \left(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}} \right) \\
&- \frac{1}{\Delta y} \left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}} \right) \\
&+ \frac{q_{z1}}{\Delta z \Delta x} \left(E^n_{x,i-\frac{1}{2},j,k+1} - E^n_{x,i-\frac{1}{2},j,k} \right) \\
&- \frac{r_{z1}}{\Delta z \Delta x} \left(E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k} \right),
\end{aligned}
\tag{3.39}
$$

and the magnetic update can be written as

$$
H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + \frac{r_{z1}}{\Delta x} \left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \right) - \frac{r_{z1}}{\Delta z} \left(E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k} \right). \tag{3.40}
$$

where

$$\alpha_{z1} = \frac{D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.41}$$

$$\beta_{z1} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z1} + \gamma_{z1}, \tag{3.42}$$

$$\gamma_{z1} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.43}$$

$$p_{z1} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{3.44}$$

$$q_{z1} = D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}, \tag{3.45}$$

$$r_{z1} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{3.46}$$

In the AMReX Yee grid, with vacuum media, the field updates are implemented as

$$
\begin{aligned}
-E_x^{n+\frac{1}{2}}(i,j-1,k) + \left(2+\frac{4\Delta y^2}{c^2\Delta t^2}\right)E_x^{n+\frac{1}{2}}(i,j,k) - E_x^{n+\frac{1}{2}}(i,j+1,k) = {} & \frac{4\Delta y^2}{c^2\Delta t^2}E_x^n(i,j,k) \\
& + \frac{2\Delta y^2}{\Delta t}\left[\frac{B_z^n(i,j,k)-B_z^n(i,j-1,k)}{\Delta y} - \frac{B_y^n(i,j,k)-B_y^n(i,j,k-1)}{\Delta z}\right] \\
& + \Delta y\left[\frac{E_y^n(i+1,j-1,k)-E_y^n(i,j-1,k)}{\Delta x} - \frac{E_y^n(i+1,j,k)-E_y^n(i,j,k)}{\Delta x}\right],
\end{aligned}
\tag{3.47a}
$$

$$
\begin{aligned}
-E_y^{n+\frac{1}{2}}(i,j,k-1) + \left(2+\frac{4\Delta z^2}{c^2\Delta t^2}\right)E_y^{n+\frac{1}{2}}(i,j,k) - E_y^{n+\frac{1}{2}}(i,j,k+1) = {} & \frac{4\Delta z^2}{c^2\Delta t^2}E_y^n(i,j,k) \\
& + \frac{2\Delta z^2}{\Delta t}\left[\frac{B_x^n(i,j,k)-B_x^n(i,j,k-1)}{\Delta z} - \frac{B_z^n(i,j,k)-B_z^n(i-1,j,k)}{\Delta x}\right] \\
& + \Delta z\left[\frac{E_z^n(i,j+1,k-1)-E_z^n(i,j,k-1)}{\Delta y} - \frac{E_z^n(i,j+1,k)-E_z^n(i,j,k)}{\Delta y}\right],
\end{aligned}
\tag{3.47b}
$$

$$
\begin{aligned}
-E_z^{n+\frac{1}{2}}(i-1,j,k) + \left(2+\frac{4\Delta x^2}{c^2\Delta t^2}\right)E_z^{n+\frac{1}{2}}(i,j,k) - E_z^{n+\frac{1}{2}}(i+1,j,k) = {} & \frac{4\Delta x^2}{c^2\Delta t^2}E_z^n(i,j,k) \\
& + \frac{2\Delta x^2}{\Delta t}\left[\frac{B_y^n(i,j,k)-B_y^n(i-1,j,k)}{\Delta x} - \frac{B_x^n(i,j,k)-B_x^n(i,j-1,k)}{\Delta y}\right] \\
& + \Delta x\left[\frac{E_x^n(i-1,j,k+1)-E_x^n(i-1,j,k)}{\Delta z} - \frac{E_x^n(i,j,k+1)-E_x^n(i,j,k)}{\Delta z}\right],
\end{aligned}
\tag{3.47c}
$$

$$
B_x^{n+\frac{1}{2}}(i,j,k) = B_x^n(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_y^{n+\frac{1}{2}}(i,j,k+1)-E_y^{n+\frac{1}{2}}(i,j,k)}{\Delta z} - \frac{E_z^n(i,j+1,k)-E_z^n(i,j,k)}{\Delta y}\right], \tag{3.47d}
$$

$$
B_y^{n+\frac{1}{2}}(i,j,k) = B_y^n(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_z^{n+\frac{1}{2}}(i+1,j,k)-E_z^{n+\frac{1}{2}}(i,j,k)}{\Delta x} - \frac{E_x^n(i,j,k+1)-E_x^n(i,j,k)}{\Delta z}\right], \tag{3.47e}
$$

$$
B_z^{n+\frac{1}{2}}(i,j,k) = B_z^n(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_x^{n+\frac{1}{2}}(i,j+1,k)-E_x^{n+\frac{1}{2}}(i,j,k)}{\Delta y} - \frac{E_y^n(i+1,j,k)-E_y^n(i,j,k)}{\Delta x}\right]. \tag{3.47f}
$$

3.2 Second ADI Half-Step

E_x and H_y By equation (1.2a),

$$
\varepsilon_{i+\frac{1}{2},j,k}\frac{E^{n+1}_{x,i+\frac{1}{2},j,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta t/2} + \sigma_{i+\frac{1}{2},j,k}\frac{E^{n+1}_{x,i+\frac{1}{2},j,k} + E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{2} = [H_4], \tag{3.48}
$$

where

$$
[H_4] = \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+1}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}. \tag{3.49}
$$

Rearranging gives

$$
\left(\frac{2\varepsilon}{\Delta t}+\frac{\sigma}{2}\right)E_x^{n+1} = \left(\frac{2\varepsilon}{\Delta t}-\frac{\sigma}{2}\right)E_x^{n+\frac{1}{2}} + [H_4]. \tag{3.50}
$$

Hence

$$
E^{n+1}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k}\,E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k}\,[H_4], \tag{3.51}
$$

By equation (1.3b),

$$H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}\Bigg(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta z}\Bigg), \tag{3.52}$$

Substituting the magnetic update into (3.51) and eliminating H_y^{n+1} terms gives

$$\begin{aligned} E^{n+1}_{x,i+\frac{1}{2},j,k} = {} & C_{a,i+\frac{1}{2},j,k}\,E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ & + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\ & - \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z}\Bigg[H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}\left(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta z}\right) \\ & \qquad - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}} - D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}\left(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k-\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k-\frac{1}{2}}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j,k} - E^{n+1}_{x,i+\frac{1}{2},j,k-1}}{\Delta z}\right)\Bigg]. \end{aligned} \tag{3.53}$$

Collecting the unknown E_x^{n+1} terms gives

$$\begin{aligned} & -\frac{C_{b,i+\frac{1}{2},j,k}\,D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2}\,E^{n+1}_{x,i+\frac{1}{2},j,k+1} \\ & + \left[1 + \frac{C_{b,i+\frac{1}{2},j,k}\,D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i+\frac{1}{2},j,k}\,D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2}\right]E^{n+1}_{x,i+\frac{1}{2},j,k} \\ & - \frac{C_{b,i+\frac{1}{2},j,k}\,D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2}\,E^{n+1}_{x,i+\frac{1}{2},j,k-1} \\ = {} & C_{a,i+\frac{1}{2},j,k}\,E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\ & - \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z}\left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right) \\ & - \frac{C_{b,i+\frac{1}{2},j,k}\,D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z\,\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right) \\ & + \frac{C_{b,i+\frac{1}{2},j,k}\,D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z\,\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k-\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k-\frac{1}{2}}\right). \end{aligned} \tag{3.54}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned} -\alpha_{x2}E^{n+1}_{x,i+\frac{1}{2},j,k-1} + \beta_{x2}E^{n+1}_{x,i+\frac{1}{2},j,k} - \gamma_{x2}E^{n+1}_{x,i+\frac{1}{2},j,k+1} = {} & p_{x2}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ & + \frac{1}{\Delta y}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\ & - \frac{1}{\Delta z}\left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right) \\ & + \frac{q_{x2}}{\Delta x\Delta z}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k-\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k-\frac{1}{2}}\right) \\ & - \frac{r_{x2}}{\Delta x\Delta z}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right), \end{aligned} \tag{3.55}$$

and the magnetic update can be written as

$$H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + \frac{r_{x2}}{\Delta x}\Big(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\Big) - \frac{r_{x2}}{\Delta z}\Big(E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}\Big). \tag{3.56}$$

where

$$\alpha_{x2} = \frac{D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2}, \tag{3.57}$$

$$\beta_{x2} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x2} + \gamma_{x2}, \tag{3.58}$$

$$\gamma_{x2} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2}, \tag{3.59}$$

$$p_{x2} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.60}$$

$$q_{x2} = D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}, \tag{3.61}$$

$$r_{x2} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{3.62}$$

E_y **and** H_z By equation (1.2b),

$$\varepsilon_{i,j+\frac{1}{2},k}\frac{E^{n+1}_{y,i,j+\frac{1}{2},k} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k}\frac{E^{n+1}_{y,i,j+\frac{1}{2},k} + E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{2} = [H_5], \tag{3.63}$$

where

$$[H_5] = \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+1}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}. \tag{3.64}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right)E^{n+1}_y = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right)E^{n+\frac{1}{2}}_y + [H_5]. \tag{3.65}$$

Hence

$$E^{n+1}_{y,i,j+\frac{1}{2},k} = C_{a,i,j+\frac{1}{2},k}E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} + C_{b,i,j+\frac{1}{2},k}[H_5], \tag{3.66}$$

By equation (1.3c),

$$\begin{aligned} H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k}\Bigg(&\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} \\ &- \frac{E^{n+1}_{y,i+1,j+\frac{1}{2},k} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta x}\Bigg), \end{aligned} \tag{3.67}$$

Substituting the magnetic update into (3.66) and eliminating H_z^{n+1} terms gives

$$\begin{aligned} E^{n+1}_{y,i,j+\frac{1}{2},k} = C_{a,i,j+\frac{1}{2},k}&E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \\ &+ \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\Big(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\Big) \\ &- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x}\Bigg[H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k}\left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n+1}_{y,i+1,j+\frac{1}{2},k} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta x}\right) \\ &\qquad - H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k} - D_{b,i-\frac{1}{2},j+\frac{1}{2},k}\left(\frac{E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n+1}_{y,i,j+\frac{1}{2},k} - E^{n+1}_{y,i-1,j+\frac{1}{2},k}}{\Delta x}\right)\Bigg]. \end{aligned} \tag{3.68}$$

Collecting the unknown E_y^{n+1} terms gives

$$
\begin{aligned}
&-\frac{C_{b,i,j+\frac{1}{2},k}D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}E^{n+1}_{y,i+1,j+\frac{1}{2},k}\\
&+\left[1+\frac{C_{b,i,j+\frac{1}{2},k}D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}+\frac{C_{b,i,j+\frac{1}{2},k}D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}\right]E^{n+1}_{y,i,j+\frac{1}{2},k}\\
&-\frac{C_{b,i,j+\frac{1}{2},k}D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}E^{n+1}_{y,i-1,j+\frac{1}{2},k}\\
=\;& C_{a,i,j+\frac{1}{2},k}E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}+\frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}}-H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right)\\
&-\frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k}-H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right)\\
&-\frac{C_{b,i,j+\frac{1}{2},k}D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x\,\Delta y}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right)\\
&+\frac{C_{b,i,j+\frac{1}{2},k}D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x\,\Delta y}\left(E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j,k}\right).
\end{aligned}
\tag{3.69}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{y2}E^{n+1}_{y,i-1,j+\frac{1}{2},k}+\beta_{y2}E^{n+1}_{y,i,j+\frac{1}{2},k}-\gamma_{y2}E^{n+1}_{y,i+1,j+\frac{1}{2},k}=\;& p_{y2}E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\\
&+\frac{1}{\Delta z}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}}-H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right)\\
&-\frac{1}{\Delta x}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k}-H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right)\\
&+\frac{q_{y2}}{\Delta y\Delta x}\left(E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j,k}\right)\\
&-\frac{r_{y2}}{\Delta y\Delta x}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right),
\end{aligned}
\tag{3.70}
$$

and the magnetic update can be written as

$$
H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k}=H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k}+\frac{r_{y2}}{\Delta y}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right)-\frac{r_{y2}}{\Delta x}\left(E^{n+1}_{y,i+1,j+\frac{1}{2},k}-E^{n+1}_{y,i,j+\frac{1}{2},k}\right).
\tag{3.71}
$$

where

$$\alpha_{y2}=\frac{D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}, \tag{3.72}$$

$$\beta_{y2}=\frac{1}{C_{b,i,j+\frac{1}{2},k}}+\alpha_{y2}+\gamma_{y2}, \tag{3.73}$$

$$\gamma_{y2}=\frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}, \tag{3.74}$$

$$p_{y2}=\frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{3.75}$$

$$q_{y2}=D_{b,i-\frac{1}{2},j+\frac{1}{2},k}, \tag{3.76}$$

$$r_{y2}=D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.77}$$

E_z **and** H_x By equation (1.2c),

$$
\varepsilon_{i,j,k+\frac{1}{2}}\frac{E^{n+1}_{z,i,j,k+\frac{1}{2}}-E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta t/2}+\sigma_{i,j,k+\frac{1}{2}}\frac{E^{n+1}_{z,i,j,k+\frac{1}{2}}+E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{2}=[H_6],
\tag{3.78}
$$

where

$$[H_6] = \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+1}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}. \tag{3.79}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_z^{n+1} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_z^{n+\frac{1}{2}} + [H_6]. \tag{3.80}$$

Hence

$$E^{n+1}_{z,i,j,k+\frac{1}{2}} = C_{a,i,j,k+\frac{1}{2}}\, E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} + C_{b,i,j,k+\frac{1}{2}}\, [H_6], \tag{3.81}$$

By equation (1.3a),

$$\begin{aligned} H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \Bigg(&\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} \\ &- \frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y} \Bigg), \end{aligned} \tag{3.82}$$

Substituting the magnetic update into (3.81) and eliminating H_x^{n+1} terms gives

$$\begin{aligned} E^{n+1}_{z,i,j,k+\frac{1}{2}} = {} & C_{a,i,j,k+\frac{1}{2}}\, E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \\ & + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}} \right) \\ & - \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \Bigg[H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} - \frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y} \right) \\ & \qquad - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}} - D_{b,i,j-\frac{1}{2},k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k}}{\Delta z} - \frac{E^{n+1}_{z,i,j,k+\frac{1}{2}} - E^{n+1}_{z,i,j-1,k+\frac{1}{2}}}{\Delta y} \right) \Bigg]. \end{aligned} \tag{3.83}$$

Collecting the unknown E_z^{n+1} terms gives

$$\begin{aligned} & - \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}\, E^{n+1}_{z,i,j+1,k+\frac{1}{2}} \\ & + \left[1 + \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} + \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} \right] E^{n+1}_{z,i,j,k+\frac{1}{2}} \\ & - \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}\, E^{n+1}_{z,i,j-1,k+\frac{1}{2}} \\ = {} & C_{a,i,j,k+\frac{1}{2}}\, E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}} \right) \\ & - \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}} \right) \\ & - \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y\, \Delta z} \left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \right) \\ & + \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y\, \Delta z} \left(E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k} \right). \end{aligned} \tag{3.84}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{z2} E^{n+1}_{z,i,j-1,k+\frac{1}{2}} + \beta_{z2} E^{n+1}_{z,i,j,k+\frac{1}{2}} - \gamma_{z2} E^{n+1}_{z,i,j+1,k+\frac{1}{2}} = p_{z2} & E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \\
&+ \frac{1}{\Delta x}\Big(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\Big) \\
&- \frac{1}{\Delta y}\Big(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\Big) \\
&+ \frac{q_{z2}}{\Delta z \Delta y}\Big(E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k}\Big) \\
&- \frac{r_{z2}}{\Delta z \Delta y}\Big(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\Big),
\end{aligned} \tag{3.85}$$

and the magnetic update can be written as

$$H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + \frac{r_{z2}}{\Delta z}\Big(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\Big) - \frac{r_{z2}}{\Delta y}\Big(E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}\Big). \tag{3.86}$$

where

$$\alpha_{z2} = \frac{D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}, \tag{3.87}$$

$$\beta_{z2} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z2} + \gamma_{z2}, \tag{3.88}$$

$$\gamma_{z2} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}, \tag{3.89}$$

$$p_{z2} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{3.90}$$

$$q_{z2} = D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}, \tag{3.91}$$

$$r_{z2} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{3.92}$$

In the AMReX Yee grid, with vacuum media, the field updates are implemented as

$$\begin{aligned}
-E_x^{n+1}(i,j,k-1) + \left(2 + \frac{4\Delta z^2}{c^2 \Delta t^2}\right) E_x^{n+1}(i,j,k) - E_x^{n+1}(i,j,k+1) = \frac{4\Delta z^2}{c^2\Delta t^2} & E_x^{n+\frac{1}{2}}(i,j,k) \\
&+ \frac{2\Delta z^2}{\Delta t}\left[\frac{B_z^{n+\frac{1}{2}}(i,j,k) - B_z^{n+\frac{1}{2}}(i,j-1,k)}{\Delta y} - \frac{B_y^{n+\frac{1}{2}}(i,j,k) - B_y^{n+\frac{1}{2}}(i,j,k-1)}{\Delta z}\right] \\
&+ \Delta z\left[\frac{E_z^{n+\frac{1}{2}}(i+1,j,k-1) - E_z^{n+\frac{1}{2}}(i,j,k-1)}{\Delta x} - \frac{E_z^{n+\frac{1}{2}}(i+1,j,k) - E_z^{n+\frac{1}{2}}(i,j,k)}{\Delta x}\right],
\end{aligned} \tag{3.93a}$$

$$\begin{aligned}
-E_y^{n+1}(i-1,j,k) + \left(2 + \frac{4\Delta x^2}{c^2 \Delta t^2}\right) E_y^{n+1}(i,j,k) - E_y^{n+1}(i+1,j,k) = \frac{4\Delta x^2}{c^2\Delta t^2} & E_y^{n+\frac{1}{2}}(i,j,k) \\
&+ \frac{2\Delta x^2}{\Delta t}\left[\frac{B_x^{n+\frac{1}{2}}(i,j,k) - B_x^{n+\frac{1}{2}}(i,j,k-1)}{\Delta z} - \frac{B_z^{n+\frac{1}{2}}(i,j,k) - B_z^{n+\frac{1}{2}}(i-1,j,k)}{\Delta x}\right] \\
&+ \Delta x\left[\frac{E_x^{n+\frac{1}{2}}(i-1,j+1,k) - E_x^{n+\frac{1}{2}}(i-1,j,k)}{\Delta y} - \frac{E_x^{n+\frac{1}{2}}(i,j+1,k) - E_x^{n+\frac{1}{2}}(i,j,k)}{\Delta y}\right],
\end{aligned} \tag{3.93b}$$

$$-E_z^{n+1}(i,j-1,k) + \left(2 + \frac{4\Delta y^2}{c^2\Delta t^2}\right) E_z^{n+1}(i,j,k) - E_z^{n+1}(i,j+1,k) = \frac{4\Delta y^2}{c^2\Delta t^2} E_z^{n+\frac{1}{2}}(i,j,k)$$
$$+ \frac{2\Delta y^2}{\Delta t}\left[\frac{B_y^{n+\frac{1}{2}}(i,j,k) - B_y^{n+\frac{1}{2}}(i-1,j,k)}{\Delta x} - \frac{B_x^{n+\frac{1}{2}}(i,j,k) - B_x^{n+\frac{1}{2}}(i,j-1,k)}{\Delta y}\right]$$
$$+ \Delta y\left[\frac{E_y^{n+\frac{1}{2}}(i,j-1,k+1) - E_y^{n+\frac{1}{2}}(i,j-1,k)}{\Delta z} - \frac{E_y^{n+\frac{1}{2}}(i,j,k+1) - E_y^{n+\frac{1}{2}}(i,j,k)}{\Delta z}\right], \tag{3.93c}$$

$$B_x^{n+1}(i,j,k) = B_x^{n+\frac{1}{2}}(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_y^{n+\frac{1}{2}}(i,j,k+1) - E_y^{n+\frac{1}{2}}(i,j,k)}{\Delta z} - \frac{E_z^{n+1}(i,j+1,k) - E_z^{n+1}(i,j,k)}{\Delta y}\right], \tag{3.93d}$$

$$B_y^{n+1}(i,j,k) = B_y^{n+\frac{1}{2}}(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_z^{n+\frac{1}{2}}(i+1,j,k) - E_z^{n+\frac{1}{2}}(i,j,k)}{\Delta x} - \frac{E_x^{n+1}(i,j,k+1) - E_x^{n+1}(i,j,k)}{\Delta z}\right], \tag{3.93e}$$

$$B_z^{n+1}(i,j,k) = B_z^{n+\frac{1}{2}}(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_x^{n+\frac{1}{2}}(i,j+1,k) - E_x^{n+\frac{1}{2}}(i,j,k)}{\Delta y} - \frac{E_y^{n+1}(i+1,j,k) - E_y^{n+1}(i,j,k)}{\Delta x}\right]. \tag{3.93f}$$

4 Solution of Linear Systems

See Tridiagonal matrix algorithm for more details.

5 Perfect Electric Conductor Boundary Conditions

For a perfect electric conductor (PEC) boundary with outward unit normal $\hat{\mathbf{n}}$, the electric field satisfies

$$\hat{\mathbf{n}} \times \mathbf{E} = 0. \tag{5.1}$$

That is, the tangential electric field vanishes on the conducting surface. If the field inside the conductor is not evolved and the initial data are consistent with the boundary, then Faraday's law also implies that the normal magnetic flux remains zero on the wall,

$$\hat{\mathbf{n}} \cdot \mathbf{B} = 0. \tag{5.2}$$

Now consider a rectangular domain with PEC boundaries on the two faces normal to x, namely the planes $x = x_{\mathrm{lo}}$ and $x = x_{\mathrm{hi}}$, while the four faces normal to y and z remain periodic. On the AMReX Yee grid, E_y and E_z are nodal in x and therefore live directly on the x-boundary planes, whereas E_x is centered at $i + \frac{1}{2}$ in the x direction and is not stored on the boundary itself. If the domain has N_x cells in x, the discrete PEC conditions are therefore

$$E_y(0,j,k) = E_y(N_x,j,k) = 0, \qquad E_z(0,j,k) = E_z(N_x,j,k) = 0. \tag{5.3}$$

No value is imposed directly on E_x at the wall, since it is the normal component.
In the Yee update, the implementation is to overwrite the tangential electric fields on the two PEC planes after each electric-field stage,

$$E_y^m(0,j,k) = E_y^m(N_x,j,k) = 0, \qquad E_z^m(0,j,k) = E_z^m(N_x,j,k) = 0, \tag{5.4}$$

where $m = n + \frac{1}{2}$ or $n+1$ depending on the stage. The neighboring magnetic updates then use differences against these zero boundary values. For example, the x-derivative appearing in the B_y update is approximated by

$$\left.\frac{\partial E_z}{\partial x}\right|_{i+\frac{1}{2},j,k} \approx \frac{E_z(i+1,j,k) - E_z(i,j,k)}{\Delta x}. \tag{5.5}$$

At the first interior face next to $x = x_{\mathrm{lo}}$,

$$\left.\frac{\partial E_z}{\partial x}\right|_{\frac{1}{2},j,k} \approx \frac{E_z(1,j,k) - E_z(0,j,k)}{\Delta x} = \frac{E_z(1,j,k)}{\Delta x}, \tag{5.6}$$

while at the last interior face next to $x = x_{\mathrm{hi}}$,

$$\left.\frac{\partial E_z}{\partial x}\right|_{N_x-\frac{1}{2},j,k} \approx \frac{E_z(N_x,j,k) - E_z(N_x-1,j,k)}{\Delta x} = -\frac{E_z(N_x-1,j,k)}{\Delta x}. \tag{5.7}$$

Likewise, the update for the normal magnetic component B_x on the PEC planes involves only tangential derivatives of E_y and E_z, so if B_x is initialized consistently on the wall then it remains zero there.
In the ADI scheme, the same PEC condition enters as a Dirichlet condition whenever a tangential electric component is solved implicitly along x. Because the y and z directions remain periodic, the implicit solves along those directions are unchanged and stay cyclic. The only modified line solves are the x-directed ones:

- the E_z solve in the first half-step, equation (3.39);
- the E_y solve in the second half-step, equation (3.70).

For the first half-step, the E_z solve uses

$$E^{n+\frac{1}{2}}_{z,0,j,k+\frac{1}{2}} = 0, \qquad E^{n+\frac{1}{2}}_{z,N_x,j,k+\frac{1}{2}} = 0. \tag{5.8}$$

Hence the first and last interior tridiagonal equations are

$$\beta_{z1} E^{n+\frac{1}{2}}_{z,1,j,k+\frac{1}{2}} - \gamma_{z1} E^{n+\frac{1}{2}}_{z,2,j,k+\frac{1}{2}} = \mathrm{RHS}, \tag{5.9}$$

and

$$-\alpha_{z1} E^{n+\frac{1}{2}}_{z,N_x-2,j,k+\frac{1}{2}} + \beta_{z1} E^{n+\frac{1}{2}}_{z,N_x-1,j,k+\frac{1}{2}} = \mathrm{RHS}, \tag{5.10}$$

respectively. The boundary unknowns do not participate in the solve; they are prescribed by the PEC condition.
Similarly, in the second half-step the E_y solve uses

$$E^{n+1}_{y,0,j+\frac{1}{2},k} = 0, \qquad E^{n+1}_{y,N_x,j+\frac{1}{2},k} = 0. \tag{5.11}$$

Hence the first and last interior tridiagonal equations are

$$\beta_{y2} E^{n+1}_{y,1,j+\frac{1}{2},k} - \gamma_{y2} E^{n+1}_{y,2,j+\frac{1}{2},k} = \mathrm{RHS}, \tag{5.12}$$

and

$$-\alpha_{y2} E^{n+1}_{y,N_x-2,j+\frac{1}{2},k} + \beta_{y2} E^{n+1}_{y,N_x-1,j+\frac{1}{2},k} = \mathrm{RHS}, \tag{5.13}$$

respectively.
Note that we do not need to explicitly impose $\hat{\mathbf{n}} \cdot \mathbf{B} = 0$ separately. For an x-normal wall, the magnetic update on the boundary plane $i = i_b$, with $i_b \in \{0, N_x\}$, is

$$B_x^{n+1}(i_b,j,k) = B_x^n(i_b,j,k) + \frac{\Delta t}{2}\left[\frac{E_y^{n+\frac{1}{2}}(i_b,j,k+1) - E_y^{n+\frac{1}{2}}(i_b,j,k)}{\Delta z} - \frac{E_z^{n+1}(i_b,j+1,k) - E_z^{n+1}(i_b,j,k)}{\Delta y}\right]. \tag{5.14}$$

Since the PEC condition gives

$$E_y(i_b,j,k) = 0, \qquad E_z(i_b,j,k) = 0 \tag{5.15}$$

for all j and k, it follows that

$$B_x^{n+1}(i_b,j,k) = B_x^n(i_b,j,k). \tag{5.16}$$

Therefore, if

$$B_x^0(i_b,j,k) = 0, \tag{5.17}$$

then

$$B_x^n(i_b,j,k) = 0 \tag{5.18}$$

is already satisfied for all n.

6 Perfect Matching Layer Boundary Conditions

Here we derive the convolutional PML boundary conditions for the ADI scheme from [2].

6.1 First ADI Half-Step

E_x **and** H_z Using the same formulation as (3.9), the PML tridiagonal system is

$$
\begin{aligned}
-\alpha_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} + \beta_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - \gamma_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} = p_{x1}E^{n}_{x,i+\frac{1}{2},j,k} \\
+ \frac{1}{\kappa_{y,i+\frac{1}{2},j,k}\Delta y}\left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\
- \frac{1}{\kappa_{z,i+\frac{1}{2},j,k}\Delta z}\left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right) \\
+ \frac{q_{x1}}{\kappa_{y,i+\frac{1}{2},j,k}\kappa_{x,i+\frac{1}{2},j-\frac{1}{2},k}\Delta x\Delta y}\left(E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}\right) \\
- \frac{r_{x1}}{\kappa_{y,i+\frac{1}{2},j,k}\kappa_{x,i+\frac{1}{2},j+\frac{1}{2},k}\Delta x\Delta y}\left(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\right) \\
- \frac{r_{x1}}{\kappa_{y,i+\frac{1}{2},j,k}\Delta y}\left(\psi^{n}_{h\,zy,i+\frac{1}{2},j+\frac{1}{2},k} - \psi^{n}_{h\,zx,i+\frac{1}{2},j+\frac{1}{2},k}\right) \\
+ \frac{q_{x1}}{\kappa_{y,i+\frac{1}{2},j,k}\Delta y}\left(\psi^{n}_{h\,zy,i+\frac{1}{2},j-\frac{1}{2},k} - \psi^{n}_{h\,zx,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\
+ \left(\psi^{n}_{e\,xz,i+\frac{1}{2},j,k} - \psi^{n}_{e\,xy,i+\frac{1}{2},j,k}\right).
\end{aligned}
\tag{6.1}
$$

Here

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\kappa_{y,i+\frac{1}{2},j,k}\kappa_{y,i+\frac{1}{2},j-\frac{1}{2},k}\Delta y^2}, \tag{6.2}$$

$$\beta_{x1} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{6.3}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\kappa_{y,i+\frac{1}{2},j,k}\kappa_{y,i+\frac{1}{2},j+\frac{1}{2},k}\Delta y^2}, \tag{6.4}$$

$$p_{x1} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{6.5}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{6.6}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{6.7}$$

and the PML magnetic update can be written as

$$
\begin{aligned}
H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + \frac{r_{x1}}{\kappa_{y,i+\frac{1}{2},j+\frac{1}{2},k}\Delta y}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right) \\
- \frac{r_{x1}}{\kappa_{x,i+\frac{1}{2},j+\frac{1}{2},k}\Delta x}\left(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\right) \\
+ r_{x1}\psi^{n}_{h\,zy,i+\frac{1}{2},j+\frac{1}{2},k} - r_{x1}\psi^{n}_{h\,zx,i+\frac{1}{2},j+\frac{1}{2},k}.
\end{aligned}
\tag{6.8}
$$

E_y **and** H_x Using the same formulation as (3.24), the PML tridiagonal system is

$$\begin{aligned}
-\alpha_{y1}E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1} + \beta_{y1}E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - \gamma_{y1}E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} = {} & p_{y1}E^{n}_{y,i,j+\frac{1}{2},k} \\
& + \frac{1}{\kappa_{z,i,j+\frac{1}{2},k}\Delta z}\left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right) \\
& - \frac{1}{\kappa_{x,i,j+\frac{1}{2},k}\Delta x}\left(H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^n_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right) \\
& + \frac{q_{y1}}{\kappa_{z,i,j+\frac{1}{2},k}\kappa_{y,i,j+\frac{1}{2},k-\frac{1}{2}}\Delta y\Delta z}\left(E^n_{z,i,j+1,k-\frac{1}{2}} - E^n_{z,i,j,k-\frac{1}{2}}\right) \\
& - \frac{r_{y1}}{\kappa_{z,i,j+\frac{1}{2},k}\kappa_{y,i,j+\frac{1}{2},k+\frac{1}{2}}\Delta y\Delta z}\left(E^n_{z,i,j+1,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}\right) \\
& - \frac{r_{y1}}{\kappa_{z,i,j+\frac{1}{2},k}\Delta z}\left(\psi^n_{h\,xz,i,j+\frac{1}{2},k+\frac{1}{2}} - \psi^n_{h\,xy,i,j+\frac{1}{2},k+\frac{1}{2}}\right) \\
& + \frac{q_{y1}}{\kappa_{z,i,j+\frac{1}{2},k}\Delta z}\left(\psi^n_{h\,xz,i,j+\frac{1}{2},k-\frac{1}{2}} - \psi^n_{h\,xy,i,j+\frac{1}{2},k-\frac{1}{2}}\right) \\
& + \left(\psi^n_{e\,yx,i,j+\frac{1}{2},k} - \psi^n_{e\,yz,i,j+\frac{1}{2},k}\right).
\end{aligned} \tag{6.9}$$

Here

$$\alpha_{y1} = \frac{D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\kappa_{z,i,j+\frac{1}{2},k}\kappa_{z,i,j+\frac{1}{2},k-\frac{1}{2}}\Delta z^2}, \tag{6.10}$$

$$\beta_{y1} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y1} + \gamma_{y1}, \tag{6.11}$$

$$\gamma_{y1} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\kappa_{z,i,j+\frac{1}{2},k}\kappa_{z,i,j+\frac{1}{2},k+\frac{1}{2}}\Delta z^2}, \tag{6.12}$$

$$p_{y1} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{6.13}$$

$$q_{y1} = D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}, \tag{6.14}$$

$$r_{y1} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{6.15}$$

and the PML magnetic update can be written as

$$\begin{aligned}
H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = {} & H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + \frac{r_{y1}}{\kappa_{z,i,j+\frac{1}{2},k+\frac{1}{2}}\Delta z}\left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\right) \\
& - \frac{r_{y1}}{\kappa_{y,i,j+\frac{1}{2},k+\frac{1}{2}}\Delta y}\left(E^n_{z,i,j+1,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}\right) \\
& + r_{y1}\psi^n_{h\,xz,i,j+\frac{1}{2},k+\frac{1}{2}} - r_{y1}\psi^n_{h\,xy,i,j+\frac{1}{2},k+\frac{1}{2}}.
\end{aligned} \tag{6.16}$$

E_z and H_y Using the same formulation as (3.39), the PML tridiagonal system is

$$
\begin{aligned}
-\alpha_{z1}E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}} + \beta_{z1}E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - \gamma_{z1}E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} = {} & p_{z1}E^{n}_{z,i,j,k+\frac{1}{2}} \\
& + \frac{1}{\kappa_{x,i,j,k+\frac{1}{2}}\Delta x}\left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right) \\
& - \frac{1}{\kappa_{y,i,j,k+\frac{1}{2}}\Delta y}\left(H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\right) \\
& + \frac{q_{z1}}{\kappa_{x,i,j,k+\frac{1}{2}}\kappa_{z,i-\frac{1}{2},j,k+\frac{1}{2}}\Delta z\Delta x}\left(E^{n}_{x,i-\frac{1}{2},j,k+1} - E^{n}_{x,i-\frac{1}{2},j,k}\right) \\
& - \frac{r_{z1}}{\kappa_{x,i,j,k+\frac{1}{2}}\kappa_{z,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta z\Delta x}\left(E^{n}_{x,i+\frac{1}{2},j,k+1} - E^{n}_{x,i+\frac{1}{2},j,k}\right) \\
& - \frac{r_{z1}}{\kappa_{x,i,j,k+\frac{1}{2}}\Delta x}\left(\psi^{n}_{h\,yx,i+\frac{1}{2},j,k+\frac{1}{2}} - \psi^{n}_{h\,yz,i+\frac{1}{2},j,k+\frac{1}{2}}\right) \\
& + \frac{q_{z1}}{\kappa_{x,i,j,k+\frac{1}{2}}\Delta x}\left(\psi^{n}_{h\,yx,i-\frac{1}{2},j,k+\frac{1}{2}} - \psi^{n}_{h\,yz,i-\frac{1}{2},j,k+\frac{1}{2}}\right) \\
& + \left(\psi^{n}_{e\,zy,i,j,k+\frac{1}{2}} - \psi^{n}_{e\,zx,i,j,k+\frac{1}{2}}\right).
\end{aligned}
\tag{6.17}
$$

Here

$$\alpha_{z1} = \frac{D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\kappa_{x,i,j,k+\frac{1}{2}}\kappa_{x,i-\frac{1}{2},j,k+\frac{1}{2}}\Delta x^2}, \tag{6.18}$$

$$\beta_{z1} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z1} + \gamma_{z1}, \tag{6.19}$$

$$\gamma_{z1} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\kappa_{x,i,j,k+\frac{1}{2}}\kappa_{x,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta x^2}, \tag{6.20}$$

$$p_{z1} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{6.21}$$

$$q_{z1} = D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}, \tag{6.22}$$

$$r_{z1} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{6.23}$$

and the PML magnetic update can be written as

$$
\begin{aligned}
H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = {} & H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + \frac{r_{z1}}{\kappa_{x,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right) \\
& - \frac{r_{z1}}{\kappa_{z,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta z}\left(E^{n}_{x,i+\frac{1}{2},j,k+1} - E^{n}_{x,i+\frac{1}{2},j,k}\right) \\
& + r_{z1}\psi^{n}_{h\,yx,i+\frac{1}{2},j,k+\frac{1}{2}} - r_{z1}\psi^{n}_{h\,yz,i+\frac{1}{2},j,k+\frac{1}{2}}.
\end{aligned}
\tag{6.24}
$$

If $\kappa_s = 1$ and all PML auxiliary variables vanish, equations (6.1), (6.9), and (6.17) reduce to (3.9), (3.24), and (3.39), respectively. Likewise, (6.8), (6.16), and (6.24) reduce to (3.10), (3.25), and (3.40).

6.2 First PML Auxiliary Variable Half-Step

The PML auxiliary variables are advanced over the first half-step from time level n to $n + \frac{1}{2}$. In this subsection, $\psi_{e\alpha s}$ denotes the electric auxiliary variable for the $\partial/\partial s$ derivative in the E_α update, and $\psi_{h\alpha s}$ denotes the magnetic auxiliary variable for the $\partial/\partial s$ derivative in the H_α update. Thus, for example, ψ_{exy} and ψ_{exz} correspond to the y- and z-derivative auxiliary variables in the E_x equation, respectively.

The electric auxiliary variables are updated from the magnetic fields at time level n,

$$\psi^{n+\frac{1}{2}}_{exy,i+\frac{1}{2},j,k} = b_{y,i+\frac{1}{2},j,k}\,\psi^{n}_{exy,i+\frac{1}{2},j,k} + a_{y,i+\frac{1}{2},j,k}\,\frac{H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^n_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y}, \tag{6.25a}$$

$$\psi^{n+\frac{1}{2}}_{exz,i+\frac{1}{2},j,k} = b_{z,i+\frac{1}{2},j,k}\,\psi^{n}_{exz,i+\frac{1}{2},j,k} + a_{z,i+\frac{1}{2},j,k}\,\frac{H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}, \tag{6.25b}$$

$$\psi^{n+\frac{1}{2}}_{eyx,i,j+\frac{1}{2},k} = b_{x,i,j+\frac{1}{2},k}\,\psi^{n}_{eyx,i,j+\frac{1}{2},k} + a_{x,i,j+\frac{1}{2},k}\,\frac{H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^n_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}, \tag{6.25c}$$

$$\psi^{n+\frac{1}{2}}_{eyz,i,j+\frac{1}{2},k} = b_{z,i,j+\frac{1}{2},k}\,\psi^{n}_{eyz,i,j+\frac{1}{2},k} + a_{z,i,j+\frac{1}{2},k}\,\frac{H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z}, \tag{6.25d}$$

$$\psi^{n+\frac{1}{2}}_{ezx,i,j,k+\frac{1}{2}} = b_{x,i,j,k+\frac{1}{2}}\,\psi^{n}_{ezx,i,j,k+\frac{1}{2}} + a_{x,i,j,k+\frac{1}{2}}\,\frac{H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x}, \tag{6.25e}$$

$$\psi^{n+\frac{1}{2}}_{ezy,i,j,k+\frac{1}{2}} = b_{y,i,j,k+\frac{1}{2}}\,\psi^{n}_{ezy,i,j,k+\frac{1}{2}} + a_{y,i,j,k+\frac{1}{2}}\,\frac{H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}. \tag{6.25f}$$

The PML recursion coefficients are,

$$b_x = \exp\left[-\left(\frac{\sigma_x}{\kappa_x} + \alpha_x\right)\frac{\Delta t}{2\varepsilon_0}\right], \tag{6.26a}$$

$$b_y = \exp\left[-\left(\frac{\sigma_y}{\kappa_y} + \alpha_y\right)\frac{\Delta t}{2\varepsilon_0}\right], \tag{6.26b}$$

$$b_z = \exp\left[-\left(\frac{\sigma_z}{\kappa_z} + \alpha_z\right)\frac{\Delta t}{2\varepsilon_0}\right], \tag{6.26c}$$

$$a_x = \frac{\sigma_x}{\kappa_x\left(\sigma_x + \kappa_x\alpha_x\right)}\left(b_x - 1\right), \tag{6.26d}$$

$$a_y = \frac{\sigma_y}{\kappa_y\left(\sigma_y + \kappa_y\alpha_y\right)}\left(b_y - 1\right), \tag{6.26e}$$

$$a_z = \frac{\sigma_z}{\kappa_z\left(\sigma_z + \kappa_z\alpha_z\right)}\left(b_z - 1\right). \tag{6.26f}$$

Here σ_s, α_s, and κ_s are evaluated at the same location as the corresponding auxiliary variable and in the same derivative direction s. For a PML layer normal to direction s, let ρ_s be the distance from the inner PML interface into the PML, $L_s = N_{\mathrm{pml}}\Delta s$ be the PML thickness, and m be the polynomial grading order. Then, for $0 \le \rho_s \le L_s$,

$$\sigma_s(\rho_s) = \sigma_{s,\max}\left(\frac{\rho_s}{L_s}\right)^m, \tag{6.27}$$

$$\kappa_s(\rho_s) = 1 + (\kappa_{s,\max} - 1)\left(\frac{\rho_s}{L_s}\right)^m, \tag{6.28}$$

$$\alpha_s(\rho_s) = \alpha_{s,\max}\left(1 - \frac{\rho_s}{L_s}\right), \tag{6.29}$$

while outside the PML,

$$\sigma_s = 0, \qquad \alpha_s = 0, \qquad \kappa_s = 1, \qquad a_s = 0. \tag{6.30}$$

The maximum conductivity may be chosen from a target reflection coefficient R as

$$\sigma_{s,\max} = -\frac{(m+1)\ln R}{2\eta_0 L_s}, \qquad \eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}}. \tag{6.31}$$

The magnetic auxiliary variables are updated from the electric fields at the time levels used by the first-half ADI magnetic updates,

$$\psi^{n+\frac{1}{2}}_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}} = b_{y,i,j+\frac{1}{2},k+\frac{1}{2}}\, \psi^{n}_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}} + a_{y,i,j+\frac{1}{2},k+\frac{1}{2}} \frac{E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta y}, \tag{6.32a}$$

$$\psi^{n+\frac{1}{2}}_{hxz,i,j+\frac{1}{2},k+\frac{1}{2}} = b_{z,i,j+\frac{1}{2},k+\frac{1}{2}}\, \psi^{n}_{hxz,i,j+\frac{1}{2},k+\frac{1}{2}} + a_{z,i,j+\frac{1}{2},k+\frac{1}{2}} \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z}, \tag{6.32b}$$

$$\psi^{n+\frac{1}{2}}_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}} = b_{x,i+\frac{1}{2},j,k+\frac{1}{2}}\, \psi^{n}_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}} + a_{x,i+\frac{1}{2},j,k+\frac{1}{2}} \frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x}, \tag{6.32c}$$

$$\psi^{n+\frac{1}{2}}_{hyz,i+\frac{1}{2},j,k+\frac{1}{2}} = b_{z,i+\frac{1}{2},j,k+\frac{1}{2}}\, \psi^{n}_{hyz,i+\frac{1}{2},j,k+\frac{1}{2}} + a_{z,i+\frac{1}{2},j,k+\frac{1}{2}} \frac{E^{n}_{x,i+\frac{1}{2},j,k+1} - E^{n}_{x,i+\frac{1}{2},j,k}}{\Delta z}, \tag{6.32d}$$

$$\psi^{n+\frac{1}{2}}_{hzx,i+\frac{1}{2},j+\frac{1}{2},k} = b_{x,i+\frac{1}{2},j+\frac{1}{2},k}\, \psi^{n}_{hzx,i+\frac{1}{2},j+\frac{1}{2},k} + a_{x,i+\frac{1}{2},j+\frac{1}{2},k} \frac{E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta x}, \tag{6.32e}$$

$$\psi^{n+\frac{1}{2}}_{hzy,i+\frac{1}{2},j+\frac{1}{2},k} = b_{y,i+\frac{1}{2},j+\frac{1}{2},k}\, \psi^{n}_{hzy,i+\frac{1}{2},j+\frac{1}{2},k} + a_{y,i+\frac{1}{2},j+\frac{1}{2},k} \frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y}. \tag{6.32f}$$

The magnetic auxiliary variables $\psi_{h\alpha s}$ are stored in the unscaled recursive-convolution form; therefore the magnetic-field coefficient D_b is applied when they enter the H updates, at the same staggered location as the corresponding magnetic field.

6.3 Second ADI Half-Step

E_x **and** H_y Using the same formulation as (3.55), the PML tridiagonal system is

$$\begin{aligned}
-\alpha_{x2}E^{n+1}_{x,i+\frac{1}{2},j,k-1} + \beta_{x2}E^{n+1}_{x,i+\frac{1}{2},j,k} - \gamma_{x2}E^{n+1}_{x,i+\frac{1}{2},j,k+1} &= p_{x2}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\
&+ \frac{1}{\kappa_{y,i+\frac{1}{2},j,k}\Delta y}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\
&- \frac{1}{\kappa_{z,i+\frac{1}{2},j,k}\Delta z}\left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right) \\
&+ \frac{q_{x2}}{\kappa_{z,i+\frac{1}{2},j,k}\kappa_{x,i+\frac{1}{2},j,k-\frac{1}{2}}\Delta x\Delta z}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k-\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k-\frac{1}{2}}\right) \\
&- \frac{r_{x2}}{\kappa_{z,i+\frac{1}{2},j,k}\kappa_{x,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta x\Delta z}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right) \\
&- \frac{r_{x2}}{\kappa_{z,i+\frac{1}{2},j,k}\Delta z}\left(\psi^{n+\frac{1}{2}}_{h\,yx,i+\frac{1}{2},j,k+\frac{1}{2}} - \psi^{n+\frac{1}{2}}_{h\,yz,i+\frac{1}{2},j,k+\frac{1}{2}}\right) \\
&+ \frac{q_{x2}}{\kappa_{z,i+\frac{1}{2},j,k}\Delta z}\left(\psi^{n+\frac{1}{2}}_{h\,yx,i+\frac{1}{2},j,k-\frac{1}{2}} - \psi^{n+\frac{1}{2}}_{h\,yz,i+\frac{1}{2},j,k-\frac{1}{2}}\right) \\
&+ \left(\psi^{n+\frac{1}{2}}_{e\,xz,i+\frac{1}{2},j,k} - \psi^{n+\frac{1}{2}}_{e\,xy,i+\frac{1}{2},j,k}\right).
\end{aligned} \tag{6.33}$$

Here

$$\alpha_{x2} = \frac{D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\kappa_{z,i+\frac{1}{2},j,k}\kappa_{z,i+\frac{1}{2},j,k-\frac{1}{2}}\Delta z^2}, \tag{6.34}$$

$$\beta_{x2} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x2} + \gamma_{x2}, \tag{6.35}$$

$$\gamma_{x2} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\kappa_{z,i+\frac{1}{2},j,k}\kappa_{z,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta z^2}, \tag{6.36}$$

$$p_{x2} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{6.37}$$

$$q_{x2} = D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}, \tag{6.38}$$

$$r_{x2} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{6.39}$$

and the PML magnetic update can be written as

$$\begin{aligned} H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} &+ \frac{r_{x2}}{\kappa_{x,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right) \\ &- \frac{r_{x2}}{\kappa_{z,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta z}\left(E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}\right) \\ &+ r_{x2}\psi^{n+\frac{1}{2}}_{h\,yx,i+\frac{1}{2},j,k+\frac{1}{2}} - r_{x2}\psi^{n+\frac{1}{2}}_{h\,yz,i+\frac{1}{2},j,k+\frac{1}{2}}. \end{aligned} \tag{6.40}$$

E_y **and** H_z Using the same formulation as (3.70), the PML tridiagonal system is

$$\begin{aligned} -\alpha_{y2}E^{n+1}_{y,i-1,j+\frac{1}{2},k} + \beta_{y2}E^{n+1}_{y,i,j+\frac{1}{2},k} - \gamma_{y2}E^{n+1}_{y,i+1,j+\frac{1}{2},k} = p_{y2}&E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \\ &+ \frac{1}{\kappa_{z,i,j+\frac{1}{2},k}\Delta z}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right) \\ &- \frac{1}{\kappa_{x,i,j+\frac{1}{2},k}\Delta x}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right) \\ &+ \frac{q_{y2}}{\kappa_{x,i,j+\frac{1}{2},k}\kappa_{y,i-\frac{1}{2},j+\frac{1}{2},k}\Delta y\Delta x}\left(E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j,k}\right) \\ &- \frac{r_{y2}}{\kappa_{x,i,j+\frac{1}{2},k}\kappa_{y,i+\frac{1}{2},j+\frac{1}{2},k}\Delta y\Delta x}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right) \\ &- \frac{r_{y2}}{\kappa_{x,i,j+\frac{1}{2},k}\Delta x}\left(\psi^{n+\frac{1}{2}}_{h\,zy,i+\frac{1}{2},j+\frac{1}{2},k} - \psi^{n+\frac{1}{2}}_{h\,zx,i+\frac{1}{2},j+\frac{1}{2},k}\right) \\ &+ \frac{q_{y2}}{\kappa_{x,i,j+\frac{1}{2},k}\Delta x}\left(\psi^{n+\frac{1}{2}}_{h\,zy,i-\frac{1}{2},j+\frac{1}{2},k} - \psi^{n+\frac{1}{2}}_{h\,zx,i-\frac{1}{2},j+\frac{1}{2},k}\right) \\ &+ \left(\psi^{n+\frac{1}{2}}_{e\,yx,i,j+\frac{1}{2},k} - \psi^{n+\frac{1}{2}}_{e\,yz,i,j+\frac{1}{2},k}\right). \end{aligned} \tag{6.41}$$

Here

$$\alpha_{y2} = \frac{D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\kappa_{x,i,j+\frac{1}{2},k}\kappa_{x,i-\frac{1}{2},j+\frac{1}{2},k}\Delta x^2}, \tag{6.42}$$

$$\beta_{y2} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y2} + \gamma_{y2}, \tag{6.43}$$

$$\gamma_{y2} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\kappa_{x,i,j+\frac{1}{2},k}\kappa_{x,i+\frac{1}{2},j+\frac{1}{2},k}\Delta x^2}, \tag{6.44}$$

$$p_{y2} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{6.45}$$

$$q_{y2} = D_{b,i-\frac{1}{2},j+\frac{1}{2},k}, \tag{6.46}$$

$$r_{y2} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{6.47}$$

and the PML magnetic update can be written as

$$\begin{aligned} H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} &+ \frac{r_{y2}}{\kappa_{y,i+\frac{1}{2},j+\frac{1}{2},k}\Delta y}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right) \\ &- \frac{r_{y2}}{\kappa_{x,i+\frac{1}{2},j+\frac{1}{2},k}\Delta x}\left(E^{n+1}_{y,i+1,j+\frac{1}{2},k} - E^{n+1}_{y,i,j+\frac{1}{2},k}\right) \\ &+ r_{y2}\psi^{n+\frac{1}{2}}_{h\,zy,i+\frac{1}{2},j+\frac{1}{2},k} - r_{y2}\psi^{n+\frac{1}{2}}_{h\,zx,i+\frac{1}{2},j+\frac{1}{2},k}. \end{aligned} \tag{6.48}$$

E_z **and** H_x Using the same formulation as (3.85), the PML tridiagonal system is

$$\begin{aligned} -\alpha_{z2}E^{n+1}_{z,i,j-1,k+\frac{1}{2}} + \beta_{z2}E^{n+1}_{z,i,j,k+\frac{1}{2}} - \gamma_{z2}E^{n+1}_{z,i,j+1,k+\frac{1}{2}} = p_{z2}&E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \\ &+ \frac{1}{\kappa_{x,i,j,k+\frac{1}{2}}\Delta x}\left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right) \\ &- \frac{1}{\kappa_{y,i,j,k+\frac{1}{2}}\Delta y}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\right) \\ &+ \frac{q_{z2}}{\kappa_{y,i,j,k+\frac{1}{2}}\kappa_{z,i,j-\frac{1}{2},k+\frac{1}{2}}\Delta z\Delta y}\left(E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k}\right) \\ &- \frac{r_{z2}}{\kappa_{y,i,j,k+\frac{1}{2}}\kappa_{z,i,j+\frac{1}{2},k+\frac{1}{2}}\Delta z\Delta y}\left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\right) \\ &- \frac{r_{z2}}{\kappa_{y,i,j,k+\frac{1}{2}}\Delta y}\left(\psi^{n+\frac{1}{2}}_{h\,xz,i,j+\frac{1}{2},k+\frac{1}{2}} - \psi^{n+\frac{1}{2}}_{h\,xy,i,j+\frac{1}{2},k+\frac{1}{2}}\right) \\ &+ \frac{q_{z2}}{\kappa_{y,i,j,k+\frac{1}{2}}\Delta y}\left(\psi^{n+\frac{1}{2}}_{h\,xz,i,j-\frac{1}{2},k+\frac{1}{2}} - \psi^{n+\frac{1}{2}}_{h\,xy,i,j-\frac{1}{2},k+\frac{1}{2}}\right) \\ &+ \left(\psi^{n+\frac{1}{2}}_{e\,zy,i,j,k+\frac{1}{2}} - \psi^{n+\frac{1}{2}}_{e\,zx,i,j,k+\frac{1}{2}}\right). \end{aligned} \tag{6.49}$$

Here

$$\alpha_{z2} = \frac{D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\kappa_{y,i,j,k+\frac{1}{2}}\kappa_{y,i,j-\frac{1}{2},k+\frac{1}{2}}\Delta y^2}, \tag{6.50}$$

$$\beta_{z2} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z2} + \gamma_{z2}, \tag{6.51}$$

$$\gamma_{z2} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\kappa_{y,i,j,k+\frac{1}{2}}\kappa_{y,i,j+\frac{1}{2},k+\frac{1}{2}}\Delta y^2}, \tag{6.52}$$

$$p_{z2} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{6.53}$$

$$q_{z2} = D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}, \tag{6.54}$$

$$r_{z2} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{6.55}$$

and the PML magnetic update can be written as

$$\begin{aligned} H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} &+ \frac{r_{z2}}{\kappa_{z,i,j+\frac{1}{2},k+\frac{1}{2}}\Delta z}\left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\right) \\ &- \frac{r_{z2}}{\kappa_{y,i,j+\frac{1}{2},k+\frac{1}{2}}\Delta y}\left(E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}\right) \\ &+ r_{z2}\psi^{n+\frac{1}{2}}_{hxz,i,j+\frac{1}{2},k+\frac{1}{2}} - r_{z2}\psi^{n+\frac{1}{2}}_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}}. \end{aligned} \tag{6.56}$$

If $\kappa_s = 1$ and all PML auxiliary variables vanish, equations (6.33), (6.41), and (6.49) reduce to (3.55), (3.70), and (3.85), respectively. Likewise, (6.40), (6.48), and (6.56) reduce to (3.56), (3.71), and (3.86).

6.4 Second PML Auxiliary Variable Half-Step

The PML auxiliary variables are advanced over the second half-step from time level $n+\frac{1}{2}$ to $n+1$. The notation for $\psi_{e\alpha s}$ and $\psi_{h\alpha s}$ is the same as in the first PML auxiliary-variable half-step.
The electric auxiliary variables are updated from the magnetic fields at time level $n+\frac{1}{2}$,

$$\psi^{n+1}_{exy,i+\frac{1}{2},j,k} = b_{y,i+\frac{1}{2},j,k}\,\psi^{n+\frac{1}{2}}_{exy,i+\frac{1}{2},j,k} + a_{y,i+\frac{1}{2},j,k}\frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y}, \tag{6.57a}$$

$$\psi^{n+1}_{exz,i+\frac{1}{2},j,k} = b_{z,i+\frac{1}{2},j,k}\,\psi^{n+\frac{1}{2}}_{exz,i+\frac{1}{2},j,k} + a_{z,i+\frac{1}{2},j,k}\frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}, \tag{6.57b}$$

$$\psi^{n+1}_{eyx,i,j+\frac{1}{2},k} = b_{x,i,j+\frac{1}{2},k}\,\psi^{n+\frac{1}{2}}_{eyx,i,j+\frac{1}{2},k} + a_{x,i,j+\frac{1}{2},k}\frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}, \tag{6.57c}$$

$$\psi^{n+1}_{eyz,i,j+\frac{1}{2},k} = b_{z,i,j+\frac{1}{2},k}\,\psi^{n+\frac{1}{2}}_{eyz,i,j+\frac{1}{2},k} + a_{z,i,j+\frac{1}{2},k}\frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z}, \tag{6.57d}$$

$$\psi^{n+1}_{ezx,i,j,k+\frac{1}{2}} = b_{x,i,j,k+\frac{1}{2}}\,\psi^{n+\frac{1}{2}}_{ezx,i,j,k+\frac{1}{2}} + a_{x,i,j,k+\frac{1}{2}}\frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x}, \tag{6.57e}$$

$$\psi^{n+1}_{ezy,i,j,k+\frac{1}{2}} = b_{y,i,j,k+\frac{1}{2}}\,\psi^{n+\frac{1}{2}}_{ezy,i,j,k+\frac{1}{2}} + a_{y,i,j,k+\frac{1}{2}}\frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}. \tag{6.57f}$$

The PML recursion coefficients are given by (6.26).

The magnetic auxiliary variables are updated from the electric fields at the time levels used by the second-half ADI magnetic updates,

$$\psi^{n+1}_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}} = b_{y,i,j+\frac{1}{2},k+\frac{1}{2}}\,\psi^{n+\frac{1}{2}}_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}} + a_{y,i,j+\frac{1}{2},k+\frac{1}{2}}\frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y}, \tag{6.58a}$$

$$\psi^{n+1}_{hxz,i,j+\frac{1}{2},k+\frac{1}{2}} = b_{z,i,j+\frac{1}{2},k+\frac{1}{2}}\,\psi^{n+\frac{1}{2}}_{hxz,i,j+\frac{1}{2},k+\frac{1}{2}} + a_{z,i,j+\frac{1}{2},k+\frac{1}{2}}\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z}, \tag{6.58b}$$

$$\psi^{n+1}_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}} = b_{x,i+\frac{1}{2},j,k+\frac{1}{2}}\,\psi^{n+\frac{1}{2}}_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}} + a_{x,i+\frac{1}{2},j,k+\frac{1}{2}}\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x}, \tag{6.58c}$$

$$\psi^{n+1}_{hyz,i+\frac{1}{2},j,k+\frac{1}{2}} = b_{z,i+\frac{1}{2},j,k+\frac{1}{2}}\,\psi^{n+\frac{1}{2}}_{hyz,i+\frac{1}{2},j,k+\frac{1}{2}} + a_{z,i+\frac{1}{2},j,k+\frac{1}{2}}\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k+1} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta z}, \tag{6.58d}$$

$$\psi^{n+1}_{hzx,i+\frac{1}{2},j+\frac{1}{2},k} = b_{x,i+\frac{1}{2},j+\frac{1}{2},k}\,\psi^{n+\frac{1}{2}}_{hzx,i+\frac{1}{2},j+\frac{1}{2},k} + a_{x,i+\frac{1}{2},j+\frac{1}{2},k}\frac{E^{n+\frac{1}{2}}_{y,i+1,j+\frac{1}{2},k} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta x}, \tag{6.58e}$$

$$\psi^{n+1}_{hzy,i+\frac{1}{2},j+\frac{1}{2},k} = b_{y,i+\frac{1}{2},j+\frac{1}{2},k}\,\psi^{n+\frac{1}{2}}_{hzy,i+\frac{1}{2},j+\frac{1}{2},k} + a_{y,i+\frac{1}{2},j+\frac{1}{2},k}\frac{E^{n+1}_{x,i+\frac{1}{2},j+1,k} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta y}. \tag{6.58f}$$

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